

# Investment Appraisal

B S Choudhary  
Department of Mining Engineering  
IIT(ISM) Dhanbad

# Contents

- ▶ Elements of investment appraisal,
- ▶ Static methods of investment appraisal,
- ▶ Dynamic methods of appraisal,
- ▶ Discounted cash flow analysis

# Investment appraisal

- ▶ Investment appraisal is a way that a business will assess the attractiveness of possible investments or projects based on the findings of several different capital budgeting and financing techniques. For traders, it is a form of fundamental analysis as it can help identify long-term trends as well as a company's perceived profitability.
- ▶ Investment appraisal is important for traders because it is a form of fundamental analysis and, as such, it is capable of showing a trader whether a stock or a company has long-term potential based on the profitability of its future projects and activities.
- ▶ If a company is involved in a number of long-term investment projects, there is also a greater risk that revenues, costs and cashflows might be damaged. This is something that a trader will need to consider before they take a position on a company's shares.
- ▶ Investment appraisal techniques are payback period, internal rate of return, net present value, accounting rate of return, and profitability index.

Investment appraisal is a collection of techniques used to identify the attractiveness of an investment. Its goals are:

- ▶ assess the viability of achieving the objectives;
- ▶ support the production of a [business case](#).

Investment appraisal is very focused on the early phases of a project or programme and is performed in parallel with the early work on [management plans](#) and [delivery plans](#).

Most investment appraisals are based on cash flows but there are other factors that may need to be included, such as:

- ▶ **Legal considerations** - a project that enables an organisation to conform to new legislation may be compulsory if the organisation is to continue to operate. An appraisal based on the return on investment is therefore less appropriate.
- ▶ **Environmental impact** - the effects of work on the natural environment are increasingly a factor when considering an investment. Environmental impact analysis of infrastructure work is written into legislation in many parts of the world.
- ▶ **Social impact** - for charitable organisations, return on investment could be measured in non-financial terms such as 'quality of life' or even 'lives saved'.
- ▶ **Operational benefits** - these could include less tangible elements such as 'increased customer satisfaction', 'higher staff morale' or 'competitive advantage'.
- ▶ **Risk** - all organisations are subject to business and operational risk. An investment decision may be justified because it reduces risk.

- ▶ **Length of the project:** The longer the project, the greater the risk that estimated revenues, costs and cash flows prove unrealistic
- ▶ **Source of the data:** Are estimated project profits and cash flows based on detailed research, emotional feel, or a little of both?
- ▶ **The size of the investment:** An investment that uses a substantial proportion of the available business funds is, by definition, more risky than a smaller project. Risk is also about the consequences to the business if something goes wrong!
- ▶ **The economic and market environment:** A major issue for most large investments. Most projects will make assumptions about demand, costs, pricing etc which can become wildly inaccurate through changing market and economic conditions
- ▶ **The experience of the management team:** A project in a market in which the management team has strong experience is a lower-risk proposition than one in which the business is taking a step into the unknown!
- ▶ **Qualitative influences on investment appraisal:** An investment decision is not just about the numbers. A spread sheet calculation for NPV or ARR might suggest a particular decision, but management also need to take account of qualitative issues such as:
  - ▶ The impact on employees
  - ▶ Product quality and customer service
  - ▶ Consistency of the investment decision with corporate objectives
  - ▶ The business' brand and image, including reputation
  - ▶ Implications for production and operations, including any disruption or change to the existing set-up
  - ▶ A business' responsibilities to society and other external stakeholders

# Basic methods of investment appraisal

## Static Methods

- Cost Comparison Method
- Profit Comparison Method
- Average Rate of Return Method
- Static Payback Period Method

## Discounted Cash Flow Methods

- Net Present Value Method
- Annuity Method
- Internal Rate of Return Method
- Dynamic Payback Period Method

# Cost comparison method (CCM)

- ▶ For the cost comparison method (CCM) the target measure is, as the name suggests, the cost(s) of an investment project. It is assumed when using the CCM that the revenues of mutually exclusive investment alternatives (and the option to forego the investment, if this is a permissible alternative) are identical and that only the costs differ. Costs analysed include: personnel expenditures (wages, salaries, social expenditure etc.), cost of raw materials, depreciation, interest, taxes and fees, and costs of outside services (such as repair or maintenance). The average costs for the planning period are determined for each investment alternative. Note that, for variable costs, the future production volume is a crucial determinant. Adding up all cost components gives a total cost for each alternative investment.
- ▶ Assessing absolute profitability on the basis of total costs is not meaningful if the revenues generated by an investment differ from those that would be generated without the investment. This is usually the case for foundational or expansion investments. In these cases, absolute profitability can be judged only on the basis of profits (i.e. not using the CCM). However, for replacement or rationalization investments, a comparison of total costs with the investment and total costs without the investment can be conclusive.
- ▶ Relative profitability can be determined using the CCM in all situations where the projects under comparison have identical revenues. When considering relative profitability, it doesn't matter what the costs would have been without the project, since simply costs between the various project options are compared.

## Key Concept:

- ▶ Absolute profitability is achieved if the total cost of making an investment is lower than the total cost of rejecting it.
- ▶ Relative profitability is achieved if making an investment results in a total cost that is lower than that of the alternative investment project(s) under consideration.

# Profit Comparison Method

- ▶ As the name suggests, the profit comparison method (PCM) differs from the cost comparison method because it considers both the cost and revenues of investment projects. The target measure is the average profit, which is determined as the difference between revenues and costs. Apart from this difference, all of the other assumptions made in the CCM continue to apply for PCM.

## Key Concept

- ▶ Absolute profitability is achieved if an investment project leads to a profit greater than zero.
- ▶ Relative profitability is achieved if an investment project leads to a higher profit than the alternative investment project(s).



The PCM is illustrated in the following example. A company has the choice between the following two investment projects A and B:

**Table 2.2** Data for the investment projects A and B (PCM)

| Data                                                  | Project A | Project B |
|-------------------------------------------------------|-----------|-----------|
| Initial investment outlay (€)                         | 180,000   | 200,000   |
| Freight charges (€)                                   | 15,000    | 25,000    |
| Set-up charges (€)                                    | 2,000     | 2,000     |
| Economic life (years)                                 | 5         | 5         |
| Liquidation value at the end of the economic life (€) | 12,000    | 17,000    |
| Other fixed costs (€ per year)                        | 4,000     | 20,000    |
| Production and sales volume (units per year)          | 9,000     | 12,000    |
| Sales price (€ per unit)                              | 10        | 10        |
| Variable costs (€ per unit)                           | 2         | 1.90      |
| Rate of interest (% per year)                         | 6         | 6         |

To assess the absolute and relative profitability of the two investment projects, the projects' average revenues and costs must be determined. The annual revenues of projects A ( $R_A$ ) and B ( $R_B$ ) amount to:

$$R_A = 9,000 \text{ units/year} \cdot \text{€}10/\text{unit} = \text{€}90,000 \text{ per year and}$$

$$R_B = 12,000 \text{ units/year} \cdot \text{€}10/\text{unit} = \text{€}120,000 \text{ per year.}$$

The average cost can be determined in the same way as for the CCM approach described in Sect. 2.1. The amounts for each cost category, as well as the average total costs of projects A ( $C_A$ ) and B ( $C_B$ ), are shown in the following table:

**Table 2.3** Cost categories for the investment projects A and B (PCM)

| Cost category (each in € per year) | Project A | Project B |
|------------------------------------|-----------|-----------|
| Depreciation                       | 37,000    | 42,000    |
| Interest                           | 6,270     | 7,320     |
| Other fixed costs                  | 4,000     | 20,000    |
| Variable costs                     | 18,000    | 22,800    |
| Total costs                        | 65,270    | 92,120    |

The average profits for alternatives A ( $P_A$ ) and B ( $P_B$ ) amount to:

$$P_A = R_A - C_A = \text{€}90,000/\text{year} - \text{€}65,270/\text{year} = \text{€}24,730 \text{ per year}$$

$$P_B = R_B - C_B = \text{€}120,000/\text{year} - \text{€}92,120/\text{year} = \text{€}27,880 \text{ per year}$$

Both investment projects achieve absolute profitability, since they earn a positive profit. Project B achieves relative profitability because of its higher average profit.

# Average Rate of Return Method

- ▶ The average rate of return (ARR) method differs from the PCM in regard to its target measure. The ARR method combines a profit measure with a capital measure to focus on the return (expressed as a rate of interest) earned on the capital invested.
- ▶ Both the profit measure and the capital measure can be defined differently. Average capital tie-up can be used as the capital measure, while the profit measure can be determined by adding average profit and average interest.
- ▶ This leads to the following formula:

$$\text{Average rate of return} = \frac{\text{Average profit} + \text{Average interests}}{\text{Average capital tie-up}}$$

The average interest is derived by applying a given interest rate to the average tie-up capital. When using the PCM, this interest is subtracted from revenues as a component of cost. For the ARR method this step is reversed by adding the interest amount back to the profit calculated using the PCM. The sum of average profit and average interest represents a surplus, which is compared to the average capital tie-up to determine the ARR method profitability measure

**Table 2.4** Relevant measures for the investment projects A and B (ARR method)

| Relevant measures          | Project A | Project B |
|----------------------------|-----------|-----------|
| Profit (€ per year)        | 24,730    | 27,880    |
| Interest (€ per year)      | 6,270     | 7,320     |
| Average capital tie-up (€) | 104,500   | 122,000   |

The average rates of return for projects A ( $ARR_A$ ) and B ( $ARR_B$ ) can be determined according to the formula given above, as follows:

$$ARR_A = \frac{€24,730/\text{year} + €6,270/\text{year}}{€104,500} = 0.2967/\text{year} \text{ (or 29.67 \%)} \text{ and}$$

$$ARR_B = \frac{€27,880/\text{year} + €7,320/\text{year}}{€122,000} = 0.2885/\text{year} \text{ (or 28.85 \%)}.$$

It is obvious that both projects achieve *absolute* profitability, since the rate of return they generate exceeds the relevant interest rate of 6 %. Investment project A achieves *relative* profitability due to its higher rate of return. This example illustrates that investment recommendations can be inconsistent between the PCM and the ARR method analyses if the considered alternatives require different capital tie-ups. An absolutely profitable investment project with lower capital tie-ups usually appears more attractive when using the ARR method than it does using the PCM.

## Accounting rate of return

- ▶ The accounting rate of return (ARR) is a ratio used in capital budgeting to calculate an investment's expected return compared to the initial cost. Unlike NPV, ARR does not account for the time value of money, and if the ARR is equal to or greater than the required rate of return, then the project is deemed to have acceptable levels of profitability.
- ▶ ARR is presented as a percentage return, meaning that an ARR of 20% means that the project is forecast to return 20p for every 100p invested over a one-year period. To calculate the ARR, you would divide the average return during a given period by the average investment in that same period.



## Accounting Rate of Return

This is usually termed as rate of return method. This method involves expressing the total expected income from the project as a percentage of its capital investment as illustrated below:

**Illustration 5.8:** Let, investment capital = ₹ 3,00,000.00  
and life of project = 5 years

### Annual income

| Year    | (₹)                |
|---------|--------------------|
| 1       | 80,000.00          |
| 2       | 1,00,000.00        |
| 3       | 1,00,000.00        |
| 4       | 1,20,000.00        |
| 5       | 1,20,000.00        |
| Total = | <u>5,20,000.00</u> |

Total income forecast = ₹ 5,20,000.00

Less: investment 100% = ₹ 3,00,000.00

Net income = ₹ 2,20,000.00

$$\text{Average net return per year} = \frac{220000}{5} = ₹ 44000.00$$

$$\text{Rate of return} = \frac{44000}{300000} \times 100 = 14.7\%$$

This investment appraisal criterion became popular because of the following advantages:

- (i) It is simple to calculate
- (ii) It is based on accounting information which is readily available
- (iii) It considers benefits over the entire life of the project

However, its shortcomings are as follows:

- (i) It is based upon accounting profit not cash flow.
- (ii) It also does not take into account the time value of money.

# Discounted Cash Flow Methods

There are numerous ways through which a business can carry out investment appraisals, but here are three of the most common techniques:

## Net present value

- ▶ Net present value (NPV) is the difference between the current value of cash inflows and the current value of cash outflows over a determined length of time. NPV is used to calculate the estimated profitability of a project and it is a form of capital budgeting which accounts for the time value of money.
- ▶ The time value of money is the principle that money is worth more in the present than an equivalent amount will be in the future because it has longer to earn interest. Cash inflows and outflows are adjusted according to the principle of the time value of money, taking available interest rates into account.
- ▶ As a result, NPV determines whether it is more financially prudent to invest in a project, or to accept a different rate of return elsewhere based on projected future returns. To calculate the NPV, you would subtract the current value of invested cash from the current value of the expected cash flows.
- ▶ If the NPV is positive, then it indicates that a project's predicted earnings or profits are greater than the anticipated costs. If the NPV is negative, then the reverse is true, and the project or investment might not be pursued by the company.

## Net Present Value (NPV)

The simplest and the most widely used DCF method for capital investment consists essentially of calculating the values in year 0 (the present) of all the cash flows, and then adding these values together. The resultant total is called the net present value of the project. If the NPV is greater than zero, the return from the project will be more than the cost of the capital, and the project should therefore, be accepted. Symbolical

$$NPV = \frac{CF_0}{(1+i)^0} + \frac{CF_1}{(1+i)^1} + \frac{CF_2}{(1+i)^2} + \dots + \frac{CF_n}{(1+i)^n}$$
$$= \sum_{t=0}^n \frac{CF_t}{(1+i)^t}$$

where,

$CF_t$  is the cash flow occurring at the end of year  $t$

$n$  is the life of the project

$i$  is the discount rate

As discussed, each cash flow is discounted to a Present Value (PV), using the cost of capital as the predetermined discount rate. The NPV measures the difference between the discounted positive cash flows and the discounted investment (negative cash flows).

To illustrate the calculation of NPV, Table 5.6 is presented:

**Table 5.6** NPV Calculation from Cash Flow

| Year  | Cash flow in (₹) | Discount rate at 10% | Value in year 0 (₹) |
|-------|------------------|----------------------|---------------------|
| 0     | -1,000           | 1                    | -1,000              |
| 1     | 200              | 0.909                | 182                 |
| 2     | 400              | 0.826                | 340                 |
| 3     | 600              | 0.751                | 451                 |
| 4     | 800              | 0.683                | 546                 |
| 5     | 1,000            | 0.621                | 621                 |
| 6     | 1,200            | 0.565                | 673                 |
| NPV = |                  |                      | 1,813               |

If NPV is greater than zero, the return from the project will be more to cover the investment, and the project should therefore, be accepted.

NPV is determined in part by the discount rate adopted. There is no unique NPV result but a distribution of results depending on the assumed cost of capital.



### Example 3.1

A company is considering an investment to expand their business. The necessary data for the two available investment options are listed in the following table:

**Table 3.1** Data for investment projects A and B (NPV method)

| Data                          | Investment project A | Investment project B |
|-------------------------------|----------------------|----------------------|
| Initial investment outlay (€) | 100,000              | 60,000               |
| Economic life (years)         | 5                    | 4                    |
| Liquidation value (€)         | 5,000                | 0                    |
| Net cash flows (€)            |                      |                      |
| t = 1                         | 28,000               | 22,000               |
| t = 2                         | 30,000               | 26,000               |
| t = 3                         | 35,000               | 28,000               |
| t = 4                         | 32,000               | 28,000               |
| t = 5                         | 30,000               | –                    |
| Relevant discount rate (%)    | 8                    | 8                    |

For the assessment of relative profitability the net present value of the alternative project B ( $NPV_B$ ) must also be calculated. It amounts to:  
 $NPV_B = €25,469.32$

It is required to determine the absolute and relative profitability of the two projects by using the NPV method.

For investment project A the net present value ( $NPV_A$ ) can be calculated as follows:

$$NPV_A = -€100,000 + €28,000 \cdot 1.08^{-1} + €30,000 \cdot 1.08^{-2} + €35,000 \cdot 1.08^{-3} + €32,000 \cdot 1.08^{-4} + €30,000 \cdot 1.08^{-5} + €5,000 \cdot 1.08^{-5}$$

$$NPV_A = €26,771.59$$

This result is interpreted as follows. Since  $NPV_A$  is positive it can be deduced that investment project A is profitable in *absolute* terms. The NPV of €26,771.59 is the net monetary gain achieved if investment project A is undertaken. This assumes a discount rate of 8 % for the interest (payments). This result would be achieved regardless of the source of finance: from internal funds, debt financing or any combination of these.

## Payback period

- ▶ Payback period is the length of time between making an investment and the time at which that investment has broken even.
- ▶ To calculate the payback period, take the cost of the investment and divide it by the annual [cash flow](#). Investments with shorter payback periods are more desirable because it will take less time for an investor to receive back their capital. The main advantages are

- (i) It is simple both in concept and application.
- (ii) It is a rough and ready method for dealing with risk.
- (iii) Since it emphasizes earlier cash inflows, it may be a sensible criterion when the firm is having problems of liquidity.

However, there are following serious limitations:

- (i) It fails to consider time value of money. Cash inflows in the payback calculation are simply added without suitable discounting.
- (ii) It ignores cash flows beyond the payback period. This leads to discrimination against projects, which generates, substantial cash inflows in later years.
- (iii) It is a measure of projects capital recovery, not profitability.

**Illustration 5.7:** For payback method, let

|                    |           |
|--------------------|-----------|
| Initial investment | ₹ 6000.00 |
| Revenue per year   | ₹ 4000.00 |
| Expenses per year  | ₹ 2500.00 |
| Profit per year    | ₹ 1500.00 |

$$\text{Payback period} = \frac{6000}{1500} = 4 \text{ years}$$

Some financial institutions use what accountants consider conventional methods of project appraisal. These methods are generally inadequate approximates compared to discounting methods, and can result in sub-optimal investment decisions. However, they are used, and project appraisers should be aware of their existence.

# Internal Rate of Return Method

- ▶ The internal rate of return represents the interest earned on the capital employed at specific points in time by the investment project under consideration.
- ▶ Absolute profitability is achieved if an investment project's internal rate of return is higher than the uniform discount rate.
- ▶ The internal rate of return is the rate that leads to a NPV of zero when applied as the uniform discount rate
- ▶ The sum of all net cash flows in the economic life is higher than or equal to zero:

$$\sum_{t=0}^T \text{NCF}_t \geq 0$$



## Internal Rate of Return or DCFROR

When the evaluators in the mineral industry speak of a rate of return on an investment proposal, they are almost always referring to the so called Discounted Cash Flow Rate of Return (DCFROR). This term is a special version of the more generic term Internal Rate of return (IRR). This criterion is employed more in the minerals industry for investment proposal evaluation than other techniques.

NPV depends on the knowledge of an external interest rate for its application (i.e., external to the project such as, firm's cost of capital). The IRR method is closely related to the NPV method as it also is a discounted cash flow method, but it seeks to avoid the arbitrary choice of an interest rate. Instead it attempts to find some interest rate initially unknown, that is, internal to the project. The procedure is to find an interest rate that will make the present value of the cash flow stream of the project to zero, that is, some interest rate that will cause the present value of cash inflows to equal the present value of cash outflows. Mathematically, IRR is defined as the interest rate that will cause the net present value to become zero.

Consider a project with initial capital outlay of ₹100 and incomes of ₹60 and ₹55 in years 1 and 2, respectively. Then, the NPV of the project at discount rate  $i$  is given by:

$$NPV = -100 + \frac{60}{(1+i)} + \frac{55}{(1+i)^2}$$

and IRR is the value of  $i$  which satisfies,

$$0 = -100 + \frac{60}{(1+i)} + \frac{55}{(1+i)^2}$$

This gives  $i = 0.1$ , i.e., the IRR of the project is 10%. In practice, the value of IRR will not always be obvious by inspection. In most cases, IRR is best calculated by evaluating the NPV for several discount rates, and then determining the discount rate corresponding to zero NPV by graphical interpolation. This method is illustrated in Table 5.7.

**Table 5.7** NPV Calculation at Various Discount Rates

| Year      | Cash flow | Discounted cash flow at |        |        |
|-----------|-----------|-------------------------|--------|--------|
|           |           | 0%                      | 5%     | 10%    |
| 0         | -1,000    | -1,000                  | -1,000 | -1,000 |
| 1         | 100       | 100                     | 95.2   | 80.9   |
| 2         | 150       | 150                     | 136.0  | 123.9  |
| 3         | 200       | 200                     | 172.8  | 150.2  |
| 4         | 250       | 250                     | 205.7  | 170.7  |
| 5         | 300       | 300                     | 235.2  | 136.3  |
| 6         | 350       | 350                     | 261.1  | 197.4  |
| Total NPV |           | 350                     | 106    | -80.6  |

Hence, IRR = 8%.

# Dynamic Payback Period Method

- ▶ The dynamic payback period method (DPP) combines the basic approach of the static payback period method with the discounting cash flow used in the NPV model. The key measure is:

## Key Concept

- ▶ The dynamic payback period is the period after which the capital invested has been recovered by the discounted net cash inflows from the project.

The determination of the DPP involves calculating the NPV of the project as at the end of every period of its economic life. As long as this value remains negative, the payback period is not yet reached. When this value reaches zero (or becomes positive for the first time), the payback period is achieved (or exceeded). If the first non-negative value exceeds zero, then payback is achieved somewhere within that last period to be considered. The part of that period (year) which must pass before payback is achieved can be approximated by interpolation.

### Example 3.4

The determination of the DPP is illustrated for the investment example discussed in the previous sections. The following table illustrates the calculations:

**Table 3.5** Determination of the DPP

| Point in time | Net cash flows (€) | Present values of discounted net cash flows (€) | Cumulative net present values (€) |
|---------------|--------------------|-------------------------------------------------|-----------------------------------|
| t             | $NCF_t$            | $NCF_t \cdot q^{-t}$                            | $NCF_t \cdot q^{-t}$              |
| 0             | -100,000           | -100,000.00                                     | -100,000.00                       |
| 1             | 28,000             | 25,925.93                                       | -74,074.07                        |
| 2             | 30,000             | 25,720.17                                       | -48,353.90                        |
| 3             | 35,000             | 27,784.13                                       | -20,569.77                        |
| 4             | 32,000             | 23,520.96                                       | +2,951.19                         |

The table lists the net cash flows for each period, the present value of the discounted net cash flows and the cumulative NPVs. The DPP is exceeded in this example after four periods, because the cumulative NPV becomes positive. So, the DPP is achieved at a point in time somewhere between 3 and 4 years after the start of the project. To approximate the actual DPP, the following linear interpolation formula may be used. Here  $t^*$  indicates the period of the last cumulated negative net present value:

$$DPP \approx t^* + \frac{NPV_{t^*}}{NPV_{t^*} - NPV_{t^*+1}} \quad (3.20)$$

When using this interpolation it is assumed that cash flows occur in a linear fashion, i.e. evenly throughout each year. In the example above, a proportion (but not all) of year 4's discounted cash flow of €23,520.96 is required before payback is achieved. The amount required is €20,569.77 (i.e. the amount outstanding at the end of year 3). It may be assumed that this proportion of the year 4 cash flow corresponds to an equivalent proportion of time elapsed during year 4. Using this assumption, the DPP of the investment project A ( $DPP_A$ ) is approximated as:

$$DPP_A \approx 3 \text{ years} + \frac{€20,569.77}{€23,520.96} = 3.87 \text{ years}$$

For project B (see Example 3.1) the DPP may be determined in the same way. It amounts to approximately 2.78 years. In contrast to the results of the NPV calculation, project B appears to be relatively profitable. Absolute profitability depends on the designated time limit. If 4 years is used, for example, then both projects are absolutely profitable. If the time limit was set at 3 years instead, then only project B would be judged absolutely profitable using the DPP criterion.



Thank you